Measure Theoretic Probability
Lesson 4
Extending a Probability Measure from a Field: The Outer Measure

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Outline

1. Recap and the Extension Problem
2. Probability Measures on Fields
3. Infimum and Supremum
4. The Outer Measure
5. Properties of the Outer Measure
6. Summary and Outlook

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Recap: Measures and Probability Measures

Measure and probability measure

On a measurable space $(\Omega, \mathcal{F})$, a **measure** is a map $\mu : \mathcal{F} \to [0, \infty]$ with $\mu(\emptyset) = 0$ and countable additivity on pairwise disjoint sets. A **probability measure** is a measure $\mathbb{P} : \mathcal{F} \to [0, 1]$ with $\mathbb{P}(\Omega) = 1$.

Three tosses of a fair coin

$\Omega = \{HHH, HHT, \ldots, TTT\}$ has 8 equally likely outcomes, so $\mathbb{P}(\{\omega\}) = \frac{1}{8}$ for each outcome ω. For $A = \{HHH, HHT\}$, finite additivity gives

$$\mathbb{P}(A) = \mathbb{P}(\{HHH\}) + \mathbb{P}(\{HHT\}) = \tfrac{1}{8} + \tfrac{1}{8} = \tfrac{1}{4}.$$

Principle

Probabilities are specified on simple sets and **extended** to larger ones by additivity.

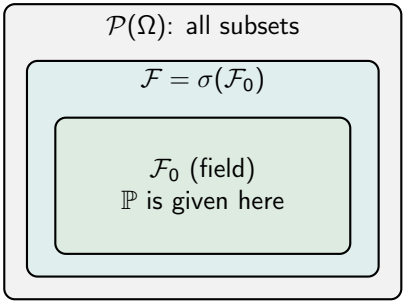
The Extension Problem

Data

- A non-empty set Ω and a **field** $\mathcal{F}_0$ of subsets of Ω (not necessarily a σ-field).
- The σ-field $\mathcal{F} = \sigma(\mathcal{F}_0)$ generated by $\mathcal{F}_0$.
- A probability measure $\mathbb{P}$ defined on $\mathcal{F}_0$ only.

Problem

Find a probability measure on $\mathcal{F}$ that **agrees with** $\mathbb{P}$ **on every set of** $\mathcal{F}_0$: new probabilities are defined, the original ones are not changed.



Plan

Define a candidate $\mathbb{P}^*$ on **all** subsets of Ω (the outer measure) and establish its basic properties. Whether $\mathbb{P}^*$ is a probability measure on $\mathcal{F}$ is the question for the next lesson.

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Fields and σ-Fields

Definition: field

A collection $\mathcal{F}_0 \subseteq \mathcal{P}(\Omega)$ is a **field** if

- $\Omega \in \mathcal{F}_0$;
- $A \in \mathcal{F}_0 \Rightarrow A^c \in \mathcal{F}_0$;
- $A, B \in \mathcal{F}_0 \Rightarrow A \cup B \in \mathcal{F}_0$.

Thus, a field is closed under finite unions and complements.

How a σ-field differs

A σ-field is also closed under **countable** unions. A field need not be: $\bigcup_{n\geq 1} A_n$ may fall outside $\mathcal{F}_0$.

Notation for this lesson

$\mathcal{F}_0$ is the given field, and $\mathcal{F} = \sigma(\mathcal{F}_0)$ is the smallest σ-field containing $\mathcal{F}_0$.

Definition: Probability Measure on a Field

Definition

A **probability measure on the field** $\mathcal{F}_0$ is a map $\mathbb{P} : \mathcal{F}_0 \to [0,1]$ such that

(P1) $\mathbb{P}(\emptyset) = 0$;

(P2) $\mathbb{P}(\Omega) = 1$;

(P3) **countable additivity, whenever it makes sense**: if $A_1, A_2, \cdots \in \mathcal{F}_0$ are pairwise disjoint **and** $\bigcup_n A_n \in \mathcal{F}_0$, then

$$\mathbb{P}\Big(\bigcup_{n=1}^{\infty} A_n\Big) = \sum_{n=1}^{\infty} \mathbb{P}(A_n).$$

Why the extra hypothesis in (P3)

Since $\mathcal{F}_0$ is only a field, $\bigcup_n A_n$ need not belong to $\mathcal{F}_0$, and then $\mathbb{P}(\bigcup_n A_n)$ is not defined. In that case (P3) says nothing.

Finite additivity still holds

Finite additivity follows by padding a finite disjoint union with $\emptyset$. Consequently, if $A \subseteq B$ and both lie in $\mathcal{F}_0$, then $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A) \geq \mathbb{P}(A)$.

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Motivation: Minimum and Maximum May Not Exist

$S = [0, 1]$

$\min S = 0$ and $\max S = 1$ both exist.

$S = (0, 1)$

- 0 is not in S, so it is not a minimum.
- Any candidate $x > 0$ is not a minimum either: $x/2$ is in S and smaller.
- Likewise there is no maximum.

Idea

Nevertheless, 0 and 1 are the "essential" minimum and maximum of $(0, 1)$ even though they are not members of the set. These are the **infimum** and the **supremum**.

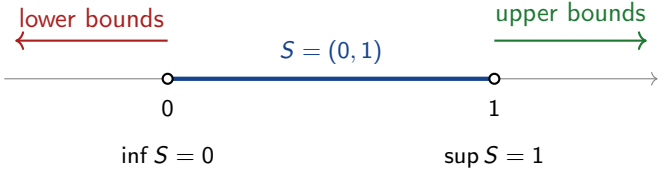
Definition: Infimum and Supremum

Infimum

A number ℓ is a **lower bound** of $S \subseteq \mathbb{R}$ if $\ell \leq s$ for all $s \in S$. The **infimum** $\inf S$ is the **greatest lower bound**: a lower bound that is $\geq$ every other lower bound.

Supremum

A number u is an **upper bound** of S if $s \leq u$ for all $s \in S$. The **supremum** $\sup S$ is the **least upper bound**.



Remark

If $\min S$ exists then $\inf S = \min S$, and if $\max S$ exists then $\sup S = \max S$. Bounds need not be strict: "lower bound" means $\leq$.

The ε-Property of Infimum and Supremum

Lemma

Let $S \subseteq \mathbb{R}$ be non-empty and $\varepsilon > 0$.

(a) If $\inf S$ is finite, there is $s \in S$ with $s < \inf S + \varepsilon$.

(b) If $\sup S$ is finite, there is $s \in S$ with $s > \sup S - \varepsilon$.

Proof of (a)

Suppose no such s exists. Then $s \geq \inf S + \varepsilon$ for all $s \in S$, so $\inf S + \varepsilon$ is a lower bound of S, larger than $\inf S$. This contradicts that $\inf S$ is the *greatest* lower bound. Part (b) is symmetric. □



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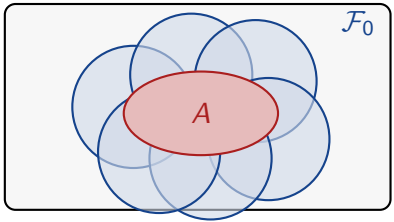
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Measuring an Arbitrary Set by Covering

Setting

Let $A \subseteq \Omega$ be **any** subset, not necessarily in $\mathcal{F}_0$ or $\mathcal{F}$. What should $\mathbb{P}(A)$ be?

- If $A \in \mathcal{F}_0$: already given.
- Otherwise **cover** A by sets $A_1, A_2, \cdots \in \mathcal{F}_0$, i.e. $A \subseteq \bigcup_n A_n$.
- The sets A_n may overlap, so the sum $\sum_n \mathbb{P}(A_n)$ overestimates the measure of the union.



Sets of $\mathcal{F}_0$ (blue) covering A.

A cover always exists

$\Omega \in \mathcal{F}_0$ and $A \subseteq \Omega$, so $A \subseteq \Omega \cup \emptyset \cup \emptyset \cup \cdots$ (finite covers are covers padded with $\emptyset$).

Definition: The Outer Measure $\mathbb{P}^*$

Definition

For every $A \subseteq \Omega$,

$$\mathbb{P}^*(A) \;=\; \inf\Big\{ \sum_{n=1}^{\infty} \mathbb{P}(A_n) \;:\; A_n \in \mathcal{F}_0 \text{ for all } n, \quad A \subseteq \bigcup_{n=1}^{\infty} A_n \Big\}.$$

Interpretation

Cover A with sets whose probabilities we know, add up the probabilities, and take the **smallest** total over all possible covers. Overlaps are counted more than once in each sum, so each sum is an over-estimate, and the infimum is the tightest one.

Remarks

- The family of covering sums is non-empty because $A \subseteq \Omega$.
- The infimum is well-defined in $[0, \infty]$; it remains to prove the measure properties.

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Property 1: $\mathbb{P}^*(\emptyset) = 0$

Proposition

$\mathbb{P}^*(\emptyset) = 0$.

Proof

Cover $\emptyset$ by the sequence $\emptyset, \emptyset, \emptyset, \ldots$ of sets of $\mathcal{F}_0$. Its sum is $\sum_n \mathbb{P}(\emptyset) = 0$. Every sum in the definition is ≥ 0, so the infimum is 0. □

Property 2: Monotonicity

Proposition

If $A \subseteq B \subseteq \Omega$, then $\mathbb{P}^*(A) \leq \mathbb{P}^*(B)$.

Proof

Every cover of B is also a cover of A (since $A \subseteq B \subseteq \bigcup_n A_n$). So the set of sums over covers of B is contained in the set of sums over covers of A. An infimum over a larger set is smaller or equal, hence

$$\mathbb{P}^*(A) \leq \mathbb{P}^*(B). \qquad \square$$

Consequence

Since $A \subseteq \Omega$, the cover $\Omega, \emptyset, \emptyset, \ldots$ gives $\mathbb{P}^*(A) \leq \mathbb{P}(\Omega) = 1$. So $\mathbb{P}^*$ takes values in $[0, 1]$.

Property 3: $\mathbb{P}^*(A) \leq \mathbb{P}(A)$ on $\mathcal{F}_0$

Proposition

If $A \in \mathcal{F}_0$, then $\mathbb{P}^*(A) \leq \mathbb{P}(A)$.

Proof

The sequence $A, \emptyset, \emptyset, \ldots$ consists of sets of $\mathcal{F}_0$ and covers A. It is one of the covers in the definition, so the infimum is at most its sum:

$$\mathbb{P}^*(A) \leq \mathbb{P}(A) + \mathbb{P}(\emptyset) + \mathbb{P}(\emptyset) + \cdots = \mathbb{P}(A). \qquad \square$$

Goal

$\mathbb{P}^*$ should **extend** $\mathbb{P}$, so we want **equality**. This is Property 5 below.

Property 4: Countable Subadditivity

Proposition

For any subsets $A_1, A_2, \ldots$ of Ω (no disjointness required),

$$\mathbb{P}^*\Big(\bigcup_{n=1}^{\infty} A_n\Big) \leq \sum_{n=1}^{\infty} \mathbb{P}^*(A_n).$$

Proof strategy

Cover each A_n almost optimally by its own sequence of $\mathcal{F}_0$-sets, with tolerance $\varepsilon/2^n$ that shrinks with n. Combining all these covers covers $\bigcup_n A_n$, and the total tolerance is $\sum_n \varepsilon/2^n = \varepsilon$.

Proof of Property 4 (1): Almost Optimal Covers

Step 1: choice of covers

Fix $\varepsilon > 0$. For each n, $\mathbb{P}^*(A_n) \leq 1$ is finite, so by the ε-property (with tolerance $\varepsilon/2^n$) there is a sequence $A_{n1}, A_{n2}, \cdots \in \mathcal{F}_0$ with

$$A_n \subseteq \bigcup_{k=1}^{\infty} A_{nk} \qquad \text{and} \qquad \sum_{k=1}^{\infty} \mathbb{P}(A_{nk}) < \mathbb{P}^*(A_n) + \frac{\varepsilon}{2^n}.$$

Step 2: sum over n

If $\sum_n \mathbb{P}^*(A_n) = \infty$, the desired inequality is immediate. Otherwise, all terms below are finite in total and

$$\sum_{n=1}^{\infty} \sum_{k=1}^{\infty} \mathbb{P}(A_{nk}) \leq \sum_{n=1}^{\infty} \mathbb{P}^*(A_n) + \varepsilon \sum_{n=1}^{\infty} \frac{1}{2^n} = \sum_{n=1}^{\infty} \mathbb{P}^*(A_n) + \varepsilon.$$

Why the sum may be split

All terms are non-negative, so the double sum can be evaluated by grouping terms. Also, $\sum_{n \geq 1} 2^{-n} = 1$.

Proof of Property 4 (2): Conclusion

Step 3: the combined family is a cover

$\bigcup_n A_n \subseteq \bigcup_n \bigcup_k A_{nk}$. The doubly indexed sets can be enumerated as one countable family in $\mathcal{F}_0$. This family covers $\bigcup_n A_n$, so by definition,

$$\mathbb{P}^*\Big(\bigcup_{n=1}^{\infty} A_n\Big) \leq \sum_{n=1}^{\infty}\sum_{k=1}^{\infty} \mathbb{P}(A_{nk}) \leq \sum_{n=1}^{\infty} \mathbb{P}^*(A_n) + \varepsilon.$$

Step 4: let $\varepsilon \to 0$

The inequality holds for every $\varepsilon > 0$, and the left side does not depend on ε. Hence

$$\mathbb{P}^*\Big(\bigcup_{n=1}^{\infty} A_n\Big) \leq \sum_{n=1}^{\infty} \mathbb{P}^*(A_n). \qquad \square$$

Note

Strict inequalities at each ε become "$\leq$" in the limit.

Property 5: $\mathbb{P}^*$ Agrees with $\mathbb{P}$ on $\mathcal{F}_0$

Lemma

Let $A \in \mathcal{F}_0$ and $A_1, A_2, \cdots \in \mathcal{F}_0$ with $A \subseteq \bigcup_n A_n$. Then $\mathbb{P}(A) \leq \sum_n \mathbb{P}(A_n)$.

Proof

Put $B_n = A \cap A_n \cap A_1^c \cap \cdots \cap A_{n-1}^c$. Then:

- $B_n \in \mathcal{F}_0$ (finitely many field operations), and $B_n \subseteq A_n$;
- the B_n are pairwise disjoint, and $\bigcup_n B_n = A \cap \bigcup_n A_n = A \in \mathcal{F}_0$.

Since the union lies in $\mathcal{F}_0$, (P3) applies: $\mathbb{P}(A) = \sum_n \mathbb{P}(B_n) \leq \sum_n \mathbb{P}(A_n)$, using monotonicity on $\mathcal{F}_0$ for each n. □

Proposition

For $A \in \mathcal{F}_0$, $\mathbb{P}^*(A) = \mathbb{P}(A)$. Indeed, the lemma shows $\mathbb{P}(A)$ is a lower bound for all covering sums, so $\mathbb{P}(A) \leq \mathbb{P}^*(A)$; together with Property 3, equality holds. In particular $\mathbb{P}^*(\Omega) = \mathbb{P}(\Omega) = 1$.

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Summary of the Construction

Properties of $\mathbb{P}^* : \mathcal{P}(\Omega) \to [0,1]$

Property	**Statement**	**Key idea**
Null set	$\mathbb{P}^*(\emptyset) = 0$	cover by $\emptyset, \emptyset, \ldots$
Monotonicity	$A \subseteq B \Rightarrow \mathbb{P}^*(A) \leq \mathbb{P}^*(B)$	covers of B cover A
Bounded by $\mathbb{P}$	$\mathbb{P}^*(A) \leq \mathbb{P}(A)$, $A \in \mathcal{F}_0$	cover $A, \emptyset, \ldots$
Countable subadditivity	$\mathbb{P}^*(\bigcup A_n) \leq \sum \mathbb{P}^*(A_n)$	$\varepsilon/2^n$ trick
Agrees with $\mathbb{P}$	$\mathbb{P}^*(A) = \mathbb{P}(A)$, $A \in \mathcal{F}_0$; $\mathbb{P}^*(\Omega) = 1$	(P3) and disjointification

Exercises

1. Find inf, sup, min, max (when they exist) of $[0,1)$, $(0,1]$ and $\{1/n : n \geq 1\}$.
2. Prove part (b) of the ε-property of the supremum.
3. Show that $\mathbb{P}^*(A \cup B) \leq \mathbb{P}^*(A) + \mathbb{P}^*(B)$ for all $A, B \subseteq \Omega$.
4. In the three-coin-toss space, let $\mathcal{F}_0 = \sigma(\{HHH\})$ with $\mathbb{P}(\{HHH\}) = \frac{1}{8}$. Compute $\mathbb{P}^*(\{HHT\})$.

Open Question and Next Lesson

What remains

$\mathbb{P}^*$ is defined on **all** subsets of Ω, takes values in $[0,1]$, satisfies $\mathbb{P}^*(\emptyset) = 0$, is monotone, countably subadditive, and extends $\mathbb{P}$. It remains to decide whether $\mathbb{P}^*$ is **countably additive** on $\mathcal{F} = \sigma(\mathcal{F}_0)$, i.e. whether it is a probability measure there.

Next lesson

Identify the sets on which $\mathbb{P}^*$ is additive, show that they include $\mathcal{F}_0$ and form a σ-field, and conclude that $\mathbb{P}^*$ defines a probability measure on $\mathcal{F} = \sigma(\mathcal{F}_0)$ extending $\mathbb{P}$.